

## Contact

dona.omondi@gmail.com

www.linkedin.com/in/don-omondi  
(LinkedIn)  
donatussss.github.io/cv/ (Personal)

## Top Skills

Electronic Engineering  
C  
KiCAD

## Certifications

Next Generation Digital Action -  
Alumni

## Honors-Awards

Best Second Year Male Student  
in Electrical and Information  
Engineering in the 2020/2021  
Academic Year

IEEEExtreme 15.0 Programming  
Competition Participation (Team:  
UoN3rdYearEEETeamA)

1st in Kenya IEEEExtreme 16.0  
Programming Competition  
Participation (Team: TheSticklers)

Acceleration Award Next Generation  
Digital Action

# Donatus Omondi

Electrical and Electronics Engineer  
Nairobi County, Kenya

## Summary

Embedded Systems Engineer with 2+ years of hands-on experience in firmware development, hardware–software integration, and PCB design for production and prototype systems. Strong background in STM32 and ESP32 platforms, real-time systems using FreeRTOS, and communication-heavy applications including CAN, Modbus, MQTT, and OCPP. Demonstrated ability to identify and correct critical design flaws, implement scalable telemetry pipelines, and deliver robust embedded solutions in EV charging, IoT tracking, and smart agriculture domains.

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## Experience

### Geviton

1 year 10 months

#### Embedded Hardware Engineer

February 2025 - Present (1 year 3 months)

Nairobi County, Kenya

- Designed and implemented BMS and charger drivers for STM32- and ESP32-based systems, enabling reliable EV battery charging and monitoring.
- Implemented OCPP-compliant firmware for EV charging infrastructure, supporting automated charging sessions and billing workflows.
- Developed telemetry and data streaming pipelines over HTTP for remote monitoring and diagnostics.
- Identified and corrected critical PCB design flaws prior to manufacturing, preventing costly board respins and resolving long-standing hardware issues.
- Developed Python packages containing custom C extensions to meet performance and portability requirements across embedded and Linux-based systems.

### Graduate Trainee

July 2024 - January 2025 (7 months)

Nairobi County, Kenya

- Developed GSM and gas sensor drivers, enabling reliable data acquisition and communication for field-deployed devices.

- Created Python-based test and certification scripts, significantly improving efficiency during mass device certification.
- Designed and implemented firmware for vehicle and motorbike trackers with remote immobilization (“kill switch”) functionality.
- Contributed firmware for remote land tracking devices used in asset and boundary monitoring.
- Reviewed and validated up to 6-layer PCB designs, issuing critical corrections before fabrication.

## Geviton

1 year

### Student Intern Embedded Firmware Engineer

October 2023 - May 2024 (8 months)

Nairobi County, Kenya

- Developed GSM, RTC, and LoRa drivers for smart farming and battery metering hardware.
- Implemented secure telemetry pipelines (MQTTs) for cloud-based data aggregation.
- Reviewed PCB designs and provided firmware-informed feedback to improve future hardware revisions.
- Extended MicroPython firmware with low-level C/C++ modules, significantly reducing power consumption and improving runtime efficiency.

## Student on Attachment

June 2023 - October 2023 (5 months)

Nairobi County, Kenya

Involved in development of embedded systems solutions.

Responsibilities include:

- PCB assembly in SMT line
- Assembly of final product in enclosures
- Firmware development using stm32 and esp32.

## Kenya Space Agency

### Student on Attachment

August 2022 - June 2023 (11 months)

Kenya

Development of the PCB design of the ADCS Subsystem of the KSA-funded Tafari 3U CubeSat

## IEEE

Student Member

October 2021 - February 2023 (1 year 5 months)

Nairobi, Kenya

## Gearbox

Student on Attachment

June 2022 - August 2022 (3 months)

Kenya

Created a flask server running on a raspberry pi on the local area network to handle user registration and image uploads. These images were used to train the AI module for face recognition access

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## Education

University of Nairobi

Undergraduate, Electrical and Electronics Engineering · (September 2019 - 2024)

Maranda High School

· (January 2015 - November 2018)

St. Hannah's Preparatory

· (May 2013 - November 2014)